

Functions Quiz:

Question 1:

What is the purpose of using functions in Python?

- a) To organize code into logical blocks
- b) To improve code readability and maintainability
- c) To enable code reuse
- d) All of the above

Question 2:

Which keyword is used to define a function in Python?

- a) `def`
- b) `function`
- c) `define`
- d) `fun`

Question 3:

Which of the following is a valid way to call a function named `my_function` with no arguments in Python?

- a) `my_function()`
- b) `call my_function()`
- c) `function my_function()`
- d) `my_function`

Question 4:

What is the scope of a variable defined inside a function in Python?

- a) Local scope
- b) Global scope
- c) Enclosing scope
- d) Built-in scope

Task 1: Add Function

Write a Python function named `add` that takes two arguments `a` and `b` and returns their sum.

Task 2: Square Function

Write a Python function named `square` that takes a number `x` as input and returns its square.

Task 3: Factorial Function

Write a Python function named `factorial` that takes a positive integer `n` as input and returns its factorial.

Task 4: Maximum Function

Write a Python function named `maximum` that takes a list of numbers as input and returns the maximum value in the list.

Task 5: Reverse Function

Write a Python function named `reverse` that takes a string `s` as input and returns its reverse.

Task 6: Check Prime Function

Write a Python function named `is_prime` that takes a positive integer `n` as input and returns `True` if `n` is prime, otherwise `False`.

Task 7: Fibonacci Function

Write a Python function named `fibonacci` that takes a positive integer `n` as input and returns the `n`th Fibonacci number.

Task 8: Palindrome Function

Write a Python function named `is_palindrome` that takes a string `s` as input and returns `True` if `s` is a palindrome, otherwise `False`.

Task 9: Sum of Squares Function

Write a Python function named `sum_of_squares` that takes a list of numbers as input and returns the sum of the squares of those numbers.

Task 10: Average Function

Write a Python function named `average` that takes a list of numbers as input and returns the average value.